

1.1 Introduction

Artificial Intelligence and Computer Vision technologies are transforming multiple industries, including education. Traditional classroom environments rely heavily on manual observation by teachers to evaluate whether students are attentive, distracted, sleepy, or actively participating during lectures. In classrooms containing a large number of students, continuous observation becomes difficult and inaccurate. As a result, many engagement-related issues remain unnoticed, directly affecting learning quality and academic performance.

The project titled “**AI-Driven Classroom Engagement Monitoring Using Computer Vision**” is designed to solve this problem through intelligent automation. The system uses Artificial Intelligence, Machine Learning concepts, OpenCV, and MediaPipe FaceMesh technology to monitor classroom engagement in real-time. The system captures video input through webcam or CCTV cameras and processes each frame continuously to identify and analyze student behavior.

The proposed system is capable of detecting multiple faces simultaneously and extracting facial landmarks using MediaPipe’s FaceMesh model. The extracted facial data is analyzed using metrics such as Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), head pose estimation, gaze direction, and facial movement tracking. Based on these calculations, the system classifies student behavior into categories such as:

- Attentive
- Sleepy
- Distracted
- Talking
- Phone Usage
- Hand Raise
- Neutral

The project also includes a modern Django-based web dashboard that displays engagement analytics, charts, percentages, and historical reports. Teachers and institutions can use this data to identify weak engagement patterns, improve teaching quality, and take corrective actions.

Unlike traditional attendance or monitoring systems, the proposed solution focuses on behavioral understanding and classroom analytics rather than only presence detection. The project demonstrates how computer vision can be effectively integrated into smart education systems.

AI-DRIVEN CLASSROOM ENGAGEMENT MONITORING SYSTEM

The complete system consists of multiple integrated modules including:

1. Video Capture Module
2. Face Detection Module
3. Behaviour Analysis Module
4. Real-Time Processing Module
5. Web Dashboard
6. Data Storage and Analytics Module

The system supports both live webcam monitoring and uploaded classroom videos, making it suitable for offline classrooms as well as online learning environments.

The project is developed using:

- Python
- Django Framework
- OpenCV
- MediaPipe
- HTML/CSS/JavaScript
- SQLite/PostgreSQL

The system aims to improve academic performance, reduce teacher workload, and enable data-driven educational analysis.

1.2 Objectives

The primary objective of the project is to develop an intelligent classroom monitoring system capable of analyzing student engagement automatically using Artificial Intelligence and Computer Vision.

The detailed objectives of the system are as follows:

1. Automated Classroom Monitoring

The system should automatically monitor classroom activities without requiring continuous manual supervision from teachers.

2. Real-Time Engagement Detection

The project aims to process video frames in real-time and identify student engagement levels continuously.

3. Behaviour Classification

The system should classify students into multiple behavioral categories such as:

- Attentive
- Sleepy
- Distracted
- Talking
- Phone Usage
- Hand Raise
- Neutral

4. Multi-Face Detection

The system should detect and track multiple student faces simultaneously using MediaPipe FaceMesh.

5. Accurate Facial Analysis

The project should analyze facial landmarks, eye movement, mouth movement, and head orientation accurately.

6. Dashboard Visualization

The system should provide a dynamic dashboard for teachers and administrators to view engagement statistics, charts, and reports.

7. Data Storage

The engagement data should be stored in a database for future analysis and reporting.

8. Teacher Assistance

The system should help teachers identify disengaged students and improve classroom interaction.

9. Scalable Architecture

The system should support future enhancements such as cloud deployment, mobile integration, and LMS integration.

10. Smart Classroom Development

The project aims to contribute toward AI-powered smart classroom environments and data-driven educational systems.

1.3 Problem Statement

Traditional classroom environments face several challenges in monitoring student engagement effectively. Teachers usually depend on manual observation to determine whether students are attentive, distracted, sleepy, or actively participating in classroom activities. In classrooms with many students, it becomes difficult for teachers to continuously monitor every individual.

Existing classroom monitoring systems mainly focus on attendance tracking and do not provide real-time behavioral analysis. Important indicators such as drowsiness, distraction, phone usage, and lack of focus often remain undetected during lectures.

The major problems identified are:

- Lack of automated classroom monitoring systems
- Difficulty in analyzing multiple students simultaneously
- Absence of real-time engagement tracking
- No historical engagement analytics
- Increased teacher workload
- Inability to identify behavioral trends
- Limited data-driven academic analysis

In online and hybrid learning environments, engagement monitoring becomes even more challenging because teachers cannot continuously observe all students through video conferencing platforms.

The proposed system solves these problems by:

- Detecting multiple student faces
- Analyzing facial behavior
- Monitoring engagement in real-time
- Generating analytics and reports
- Reducing manual observation effort
- Supporting smart educational environments

2.1 Literature Survey

Several researchers have worked on student monitoring systems and educational analytics. Existing systems mainly focus on attendance management, online learning platforms, or basic facial recognition systems.

Some systems use machine learning algorithms for emotion recognition and engagement analysis. However, many of them lack real-time processing capabilities and multi-face support. Traditional systems are also limited in identifying multiple classroom behaviors simultaneously.

The proposed system improves upon existing approaches by integrating OpenCV, MediaPipe FaceMesh, and Django into a single intelligent framework capable of real-time behavioral classification.

Modern engagement monitoring research also emphasizes privacy-preserving approaches. Researchers proposed systems that use only facial landmarks instead of storing actual student images. Such systems improve privacy protection while maintaining accurate engagement analysis.

2.1.1 Existing Research Areas

- Face detection systems
- Online attendance systems
- Emotion recognition systems
- AI-based educational analytics
- Learning management systems

2.1.2 Research Gap

Most existing systems:

- Do not support real-time classroom analysis.
- Lack of behavior classification.
- Cannot identify multiple students simultaneously.
- Do not provide live analytics dashboards.
- The proposed system addresses these limitations effectively.

AI-DRIVEN CLASSROOM ENGAGEMENT MONITORING SYSTEM

S.NO	Authors / Year	Method / System	Advantages	Disadvantages	Result
1	Mustafa Uğur Uçar & Ersin Özdemir (2022)	Real-Time Image Processing for Student Engagement Detection	Uses head pose and EAR analysis for engagement detection	Limited classroom scalability	Effective engagement scoring in online learning (MDPI)
2	Ali Abedi & Shehroz S. Khan (2021)	ResNet + TCN Hybrid Network	High accuracy using temporal analysis	Requires large datasets and GPU processing	Improved engagement detection accuracy (arXiv)
3	Prabin Sharma et al. (2019)	Eye Tracking and Head Movement Analysis	Real-time webcam-based monitoring	Limited behavioral categories	Successful engagement classification (arXiv)
4	Zibin Zhao et al. (2024)	Light_Fer Lightweight FER Model	Fast processing with fewer parameters	Limited multi-face analysis	Suitable for real-time systems (Springer)
5	Wang et al. (2024)	YOLOv5s + MediaPipe + HRV Fusion	Combines external and internal engagement factors	Complex implementation	Improved learner engagement recognition (ScienceDirect)
6	Yongfeng Qi et al. (2024)	Multimodal Vision Perspective	Integrates gaze, action, and facial analysis	High computational complexity	Accurate engagement evaluation (MDPI)
7	Farhad Mortezapour Shiri et al. (2024)	EfficientNetV2 + RNN Models	Advanced spatio-temporal classification	Requires deep learning resources	Improved online engagement prediction (Tech Science)

3.1 Introduction of Company

DharwadHubballiTutor is a professionally driven EdTech and digital solutions organization established on **15 November 2020**. The company focuses on bridging the gap between academic education and real-world industry requirements through practical training programs, technology services, and digital transformation solutions.

The organization operates at the intersection of education, software development, analytics, and digital innovation. Its primary objective is to build industry-ready talent while simultaneously helping organizations improve operational efficiency through technology-enabled services.

The company provides a wide range of services including:

- Corporate and Professional Training
- Online and Hybrid Learning Programs
- Workshops and Executive Seminars
- Software Development Solutions
- Business Intelligence Dashboards
- Data Analytics Services
- Digital Marketing Solutions

The organization serves multiple sectors including:

- Students
- Educational Institutions
- Working Professionals
- Startups
- Small Businesses
- Enterprises

The project “**AI-Driven Classroom Engagement Monitoring Using Computer Vision**” was developed under the guidance and learning ecosystem provided during the internship program.

3.2 Vision

The vision of DharwadHubballiTutor is:

“To become a trusted EdTech and digital solutions partner enabling industry-ready professionals and digitally empowered organizations.”

AI-DRIVEN CLASSROOM ENGAGEMENT MONITORING SYSTEM

The organization aims to create a technology-driven ecosystem that enhances learning quality, digital transformation, and practical skill development.

Mission

The mission of the company includes:

- Delivering industry-aligned learning solutions
- Building job-ready and future-ready talent
- Enabling organizations through software and digital services
- Bridging the gap between theory and practical implementation
- Encouraging innovation and continuous improvement

The company strongly focuses on creating practical impact through technology and education.

Objectives

The major objectives of the organization are:

1. Skill Development

To provide practical and industry-relevant technical education.

2. Technology Enablement

To help organizations adopt modern software and digital solutions.

3. Employability Improvement

To develop job-ready professionals through real-time project exposure.

4. Innovation Promotion

To encourage innovative thinking and modern problem-solving approaches.

5. Digital Transformation

To assist businesses and institutions in adapting to evolving digital technologies.

Quality Policy

The organization follows a structured quality-driven approach to ensure consistency, performance, and learner satisfaction.

The quality policy includes:

- Delivering industry-standard training programs
- Maintaining transparency and accountability
- Ensuring practical implementation-focused learning
- Continuous curriculum and technology updates
- Encouraging ethical and professional practices
- Maintaining high service delivery standards

The organization continuously evaluates performance metrics and learner outcomes to improve training quality and service efficiency.

3.3 Company Profile and Organizational Structure

Company Profile

Particular	Details
Company Name	DharwadHubballiTutor
Establishment Date	15 November 2020
Industry Type	EdTech and Digital Solutions
Services	Training, Software Development, Analytics, Digital Marketing
Focus Areas	Education, AI, Data Analytics, Digital Transformation
Target Audience	Students, Professionals, Enterprises
Working Model	Online and Hybrid
Internship Domain	Software Development and AI Systems

Leadership Team

Founder – Athar Shaikh

The founder established the organization with a vision to create an industry-oriented educational ecosystem focused on practical learning and employability enhancement. The founder has strong expertise in technology training and digital solutions.

CEO – Hifza Shaikh

The CEO manages organizational strategy, quality standards, operational scalability, and learner success initiatives. The leadership emphasizes innovation, execution excellence, and sustainable growth.

Organizational Structure

The organizational structure of the company consists of:

- Founder
- CEO
- Technical Development Team
- Training and Academic Team
- Digital Marketing Team
- Analytics and BI Team
- Operations and Support Team
- Internship and Student Development Team

The structure ensures smooth coordination between educational services and technology solutions.

3.4 Products and Services

The organization provides multiple educational and technology services.

1. Corporate and Professional Training

The company conducts technical and professional training programs focused on industry-relevant skills including:

- Python Programming
- Web Development
- Artificial Intelligence
- Data Analytics
- Software Development

2. Online and Hybrid Learning Programs

Flexible learning models are provided for students and professionals through:

- Online classes
- Hybrid training sessions
- Recorded learning materials
- Live project mentoring

3. Workshops and Executive Seminars

The company organizes workshops and seminars on:

- AI and Machine Learning
- Business Intelligence
- Software Technologies
- Career Development
- Digital Transformation

4. Software Development Services

The organization develops custom software solutions including:

- Web Applications
- Dashboard Systems
- Management Systems
- Analytics Platforms

5. BI Dashboards and Analytics

The company provides:

- Data visualization systems
- Reporting dashboards
- Business analytics tools
- Decision support systems

6. Digital Marketing Solutions

Services include:

- SEO optimization
- Social media marketing
- Brand positioning
- Growth strategy implement

3.5 SWOT Analysis

Strengths

- Strong focus on practical learning
- Industry-oriented training methods
- Modern technology adoption
- Skilled leadership and management
- Flexible online and hybrid learning models
- Expertise in AI and analytics

Weaknesses

- Limited physical infrastructure
- Growing brand recognition phase
- Dependency on digital delivery platforms

Opportunities

- Increasing demand for EdTech platforms
- Expansion into AI and analytics services
- Corporate training market growth
- Remote learning adoption
- Digital transformation demand

Threats

- High competition in EdTech sector
- Rapid technological changes
- Changing industry requirements
- Cybersecurity and data privacy challenges

3.8 Internship Contribution to the Project

During the internship period at DharwadHubballiTutor, practical exposure was gained in:

- AI-powered software systems
- Django web development
- Computer Vision concepts
- OpenCV implementation
- MediaPipe integration
- Dashboard development
- Real-time analytics systems

The internship environment helped in successfully designing and developing the project:

“AI-Driven Classroom Engagement Monitoring Using Computer Vision”

The organization provided technical guidance, industry exposure, and a practical development ecosystem for completing the project successfully.

4.1 Hardware Requirements

1. Client Side (User Devices)

The client-side system refers to devices used by teachers, administrators, or institutions to access the web dashboard and monitoring interface.

Requirements:

- **Processor:** Intel Core i3 / AMD Ryzen 3 or above
- **RAM:** Minimum 4 GB (8 GB Recommended)
- **Storage:** Minimum 128 GB Free Space
- **Browser:** Google Chrome, Mozilla Firefox, Microsoft Edge
- **Display:** 720p or higher resolution
- **Internet:** Stable broadband or Wi-Fi connection
- **Camera:** HD Webcam / CCTV Camera (720p or 1080p Recommended)

2. Server Side (Processing System)

The server-side system performs AI-based video processing, face detection, behavior classification, analytics generation, and database operations.

Requirements:

- **Processor:** Apple Silicon M4
- **RAM:** 16 GB Recommended
- **GPU:** Integrated 8-Core GPU
- **Storage:** 256 GB SSD
- **Operating System:** macOS
- **Internet:** High-speed internet connection for dashboard synchronization

4.2 Software Requirements

Operating System

- macOS

Backend

- Python 3.10+
- Django 5.x

Frontend

- HTML5
- CSS3
- JavaScript
- Chart.js

AI Libraries

- OpenCV
- MediaPipe
- NumPy
- Matplotlib

Database

- SQLite
- PostgreSQL

4.3 Introduction about Software

1. Operating System (macOS)

The system supports Windows, Linux, and macOS operating systems, making it flexible for development and usage across different platforms.

2. Frontend (HTML, CSS, JavaScript)

HTML, CSS, and JavaScript are used to develop the user interface, dashboard, analytics pages, and responsive web components.

3. Backend (Python – Django)

Python with Django framework is used for backend development, server-side processing, authentication, video handling, and database management.

4. Development Tools (Visual Studio Code)

Visual Studio Code is used for coding, debugging, testing, and managing the project efficiently.

5. Database (SQLite / PostgreSQL)

SQLite is used during development to store engagement records, user data, reports, and session details. PostgreSQL can be used for large-scale deployment.

6. AI & Computer Vision Libraries (OpenCV & MediaPipe)

OpenCV is used for video processing and frame analysis, while MediaPipe FaceMesh is used for facial landmark detection and behavior tracking.

7. Supporting Libraries (NumPy & Matplotlib)

NumPy is used for numerical calculations, and Matplotlib is used for generating graphs and engagement analytics.

8. Web Browsers (Google Chrome / Firefox / Edge)

Modern browsers are used to access the dashboard and support all frontend functionalities smoothly.

9. Version Control (Git & GitHub)

Git and GitHub are used for source code management, project backup, and version control.

10. Server Environment

Django's built-in server is used during development, and deployment can be done using Gunicorn and Nginx servers.

AI-DRIVEN CLASSROOM ENGAGEMENT MONITORING SYSTEM

5.1 Software Used

The project is developed using Python and Django framework. OpenCV is used for video processing while MediaPipe FaceMesh is used for facial landmark detection. Visual Studio Code is used as the development environment.

Software / Tool	Purpose
Python 3.10+	Core programming language used for backend and AI processing
Django Framework	Web framework used for application development
OpenCV	Video and image processing library
MediaPipe FaceMesh	Facial landmark and attention detection
Visual Studio Code (VS Code)	Main code editor and development environment
HTML, CSS, JavaScript	Frontend development
Bootstrap	Responsive user interface design
Chart.js	Analytics and graph visualization
Git & GitHub	Version control and project management
pip	Python package management
Virtual Environment (venv)	Dependency and environment management

5.2 Server

The application uses Django's built-in development server during development and testing. Future deployment can use Nginx and Gunicorn.

Django Development Server: The development server allows developers to test the application locally before deployment.

Future Deployment Server: For production deployment, the application can use more scalable and secure server technologies.

5.3 Database

SQLite is used as the primary database during development. PostgreSQL can be integrated for large-scale deployment.

The database stores:

- User information
- Uploaded videos
- Engagement analytics
- Heatmap data

6.1 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation that shows how data moves through a system. It explains the flow of information between input, processing, storage, and output components. DFD helps in understanding the working of the system clearly and systematically.

In the **AI-Driven Classroom Engagement Monitoring Using Computer Vision** system, the DFD shows how classroom video data is captured, processed, analyzed, stored, and displayed through the dashboard. The system accepts video input through webcam or CCTV cameras and processes the frames using OpenCV and MediaPipe FaceMesh.

The system analyzes facial landmarks, eye movement, head pose, and mouth movement to classify student behavior into categories such as Attentive, Sleepy, Distracted, Talking, Phone Usage, Hand Raise, and Neutral.

Symbols

External Entities

External entities are the sources or destinations of data outside the system. In this project, external entities include:

- Students
- Teachers
- Administrators
- Webcam / CCTV Camera

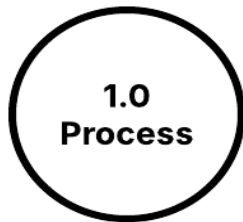


Processes

Processes transform input data into useful output. Major processes in this system include:

- Video Capture
- Face Detection

- Behaviour Analysis
- Engagement Classification
- Report Generation



Data Stores

Data stores are used to save important system data such as:

- Student engagement records
- Uploaded videos
- Analytics reports
- Session details



OR



Data Flow

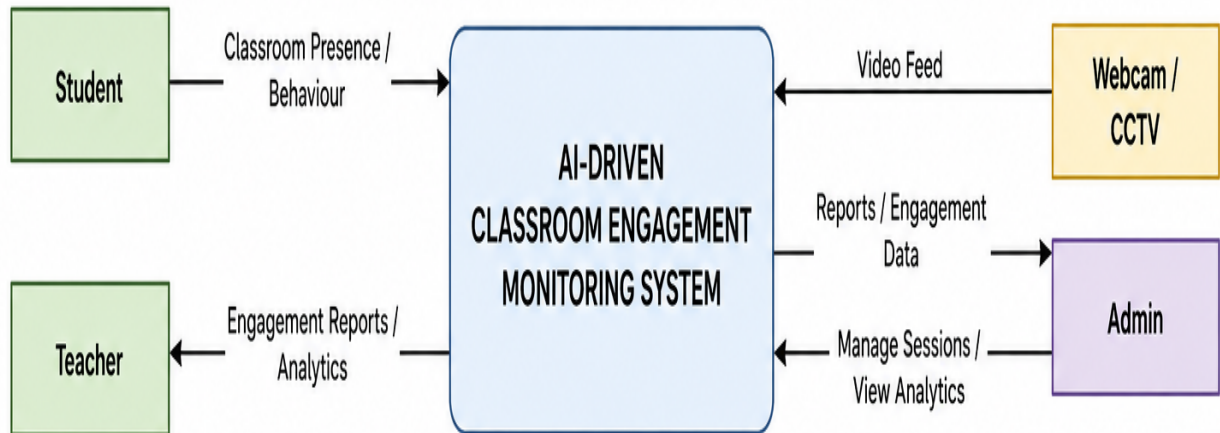
Data flows from webcam or video input to AI processing modules, then to the database and dashboard where reports and analytics are displayed.



AI-DRIVEN CLASSROOM ENGAGEMENT MONITORING SYSTEM

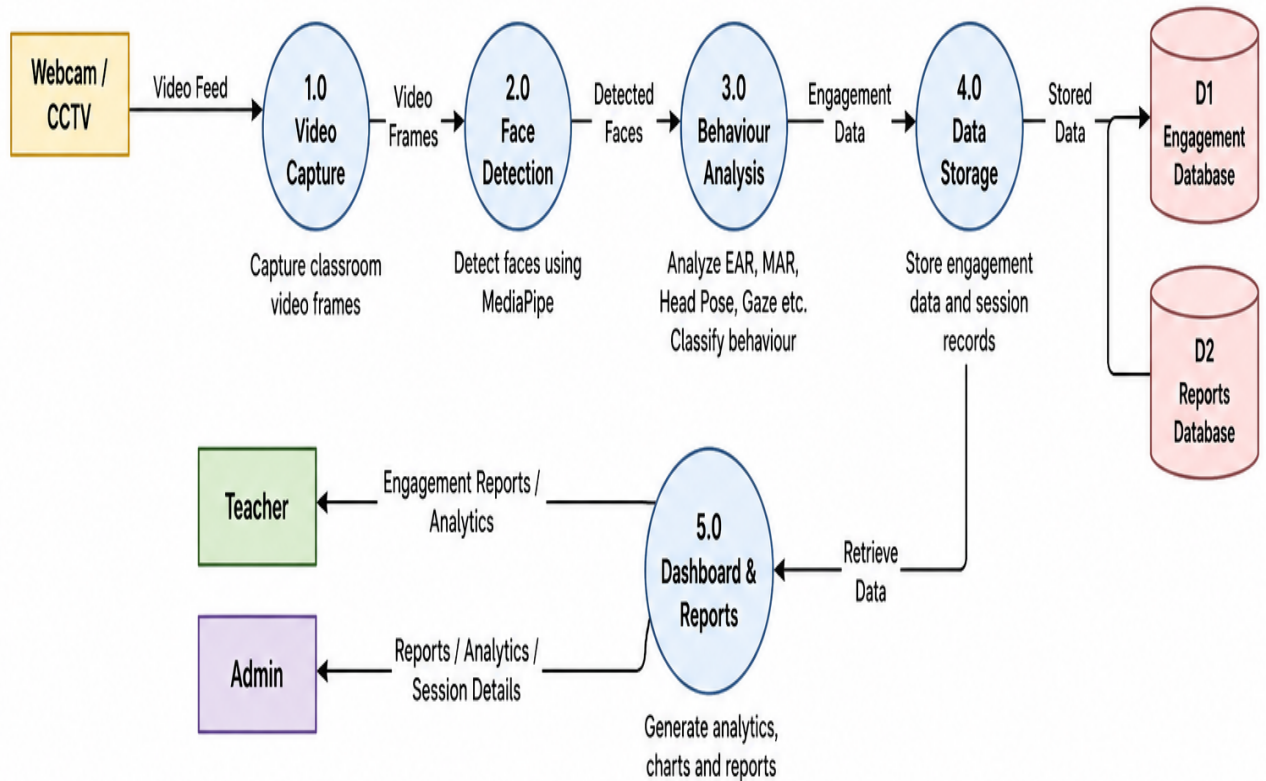
DFD- LEVEL 0

LEVEL 0 DFD (CONTEXT DIAGRAM)



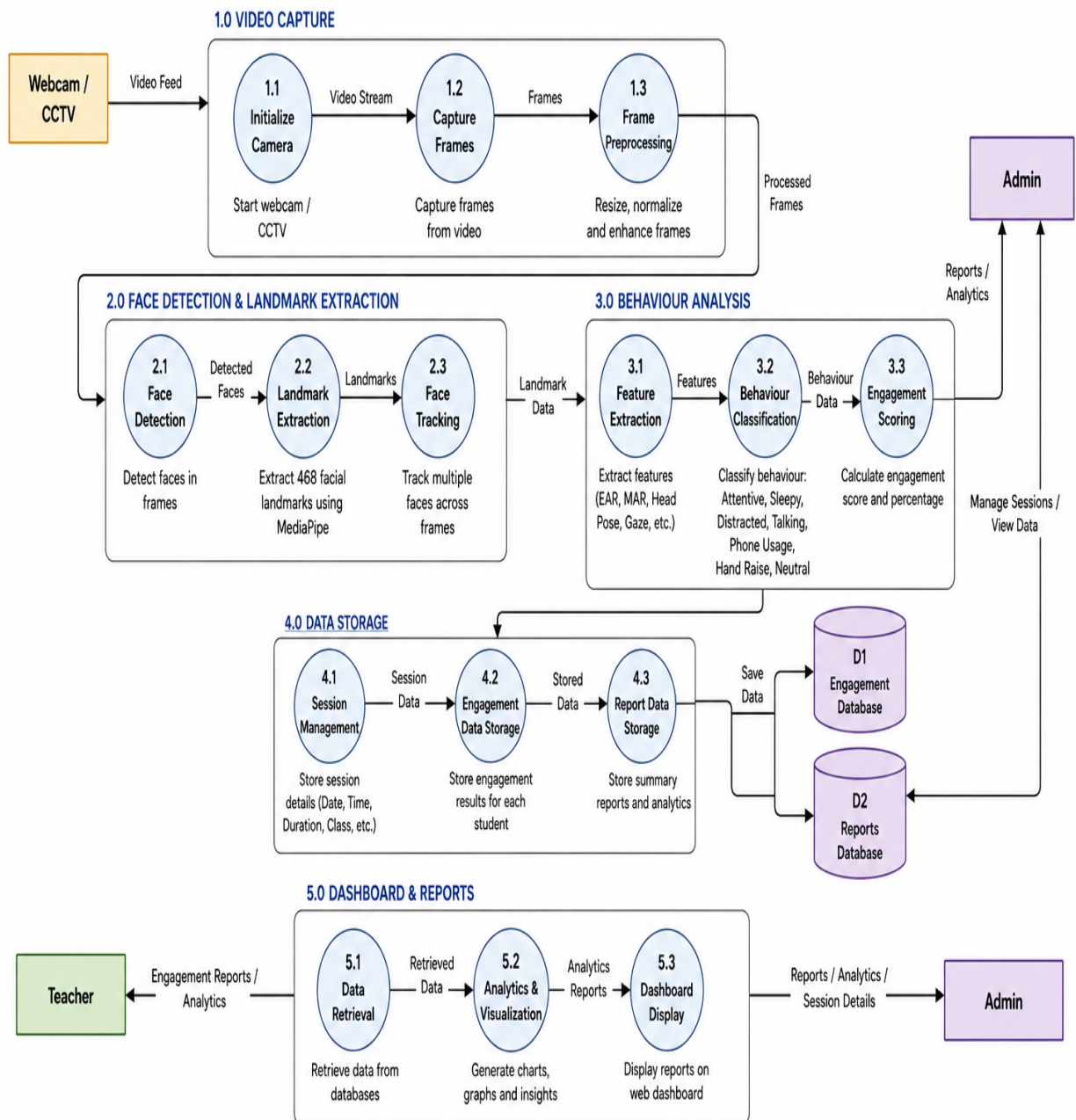
DFD-LEVEL 1

LEVEL 1 DFD



DFD-LEVEL 2

LEVEL 2 DFD – AI-DRIVEN CLASSROOM ENGAGEMENT MONITORING SYSTEM



6.2 Entity-Relationship (ER) Diagram

An Entity Relationship (ER) Diagram is a graphical representation used to design the database structure of a system. It shows the entities, attributes, and relationships between different modules of the project. ER diagrams help in understanding how data is stored, connected, and managed efficiently.

In the **AI-Driven Classroom Engagement Monitoring Using Computer Vision** system, the ER Diagram represents important entities such as Student, Classroom Session, Behaviour Analysis, Engagement Report, Admin, and Video Input. The diagram helps in organizing engagement records, session details, analytics reports, and behavioural classification data in a structured database system.

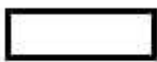
The ER Diagram also shows how students are monitored during classroom sessions, how engagement data is analyzed, and how reports are generated for teachers and administrators.

Components of ER Diagram

1. Entities

Entities are the main objects of the system. The major entities in this project are:

- Student
- Classroom Session
- Video Input
- Behaviour Analysis
- Engagement Report
- Admin



Represents Entity

These entities are represented using **rectangles**.

2. Attributes

Attributes are the properties of entities.

Example:

Student

- student_id
- student_name
- class_section
- engagement_score

Behaviour Analysis

- behaviour_type
- attention_level
- timestamp

Engagement Report

- report_id
- session_date
- analytics_data



Represents Attribute

Attributes are represented using **ovals (ellipses)**.

3. Relationships

Relationships show how entities are connected with each other.

Examples:

- Students participate in classroom sessions
- Video Input is processed for Behaviour Analysis
- Behaviour Analysis generates Engagement Reports
- Admin monitors reports and analytics



Represents Relationship

Relationships are represented using **diamonds**.

ER Diagram Notations

Weak Entity

A weak entity is an entity that cannot be uniquely identified using its own attributes alone and depends on another entity for identification. It is represented using a double rectangle.



Represents Weak Entity

Attribute

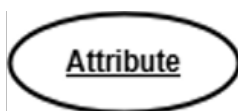
An attribute describes the properties of an entity. In ER Diagrams, attributes are represented using ovals connected to their respective entities.



Represents Attribute

Key Attribute

A key attribute uniquely identifies each record in an entity. It is represented using an underlined oval in the ER Diagram.



Attribute(s) that uniquely identify an entity instance

Multi-Valued Attribute

A multi-valued attribute can contain more than one value for a single entity. It is represented using a double oval.



Represents Multivalued Attributes

Derived Attribute

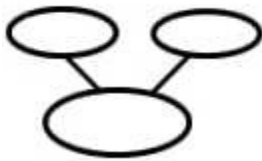
A derived attribute is calculated from another attribute and is represented using a dotted oval.



Represents Derived Attributes

Composite Attribute

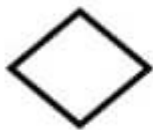
A composite attribute contains multiple smaller attributes within it. It is represented by branching sub-attributes from a main attribute.



Represents Composite Attributes

Relationship

A relationship defines the association between two or more entities in the database system. It is represented using a diamond shape.



Represents Relationship

Identifying Relationship

An identifying relationship connects a weak entity with a strong entity and is represented using a double diamond.



Represents Weak Relationships

Total Participation

Total participation indicates that every entity must participate in a relationship. It is represented using a double line.



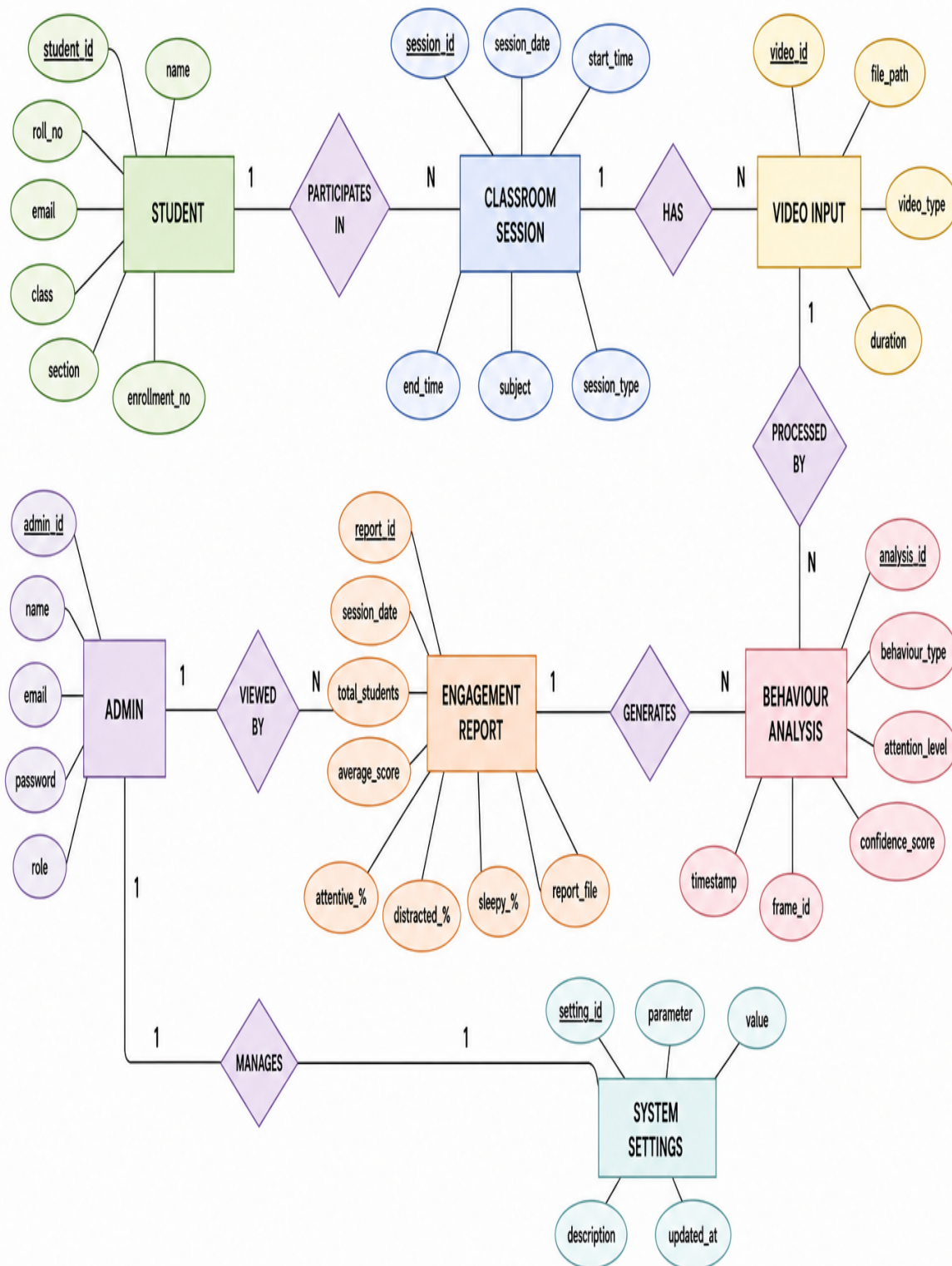
Represents Total Participation of Entity

Cardinality

Cardinality specifies how many instances of one entity are related to another entity. Common cardinalities include:

- One-to-One (1:1)
- One-to-Many (1:N)
- Many-to-One (N:1)
- Many-to-Many (M:N)

ER DIAGRAM - AI-DRIVEN CLASSROOM ENGAGEMENT MONITORING SYSTEM



6.3 Table Design

Definition

Table design is the process of organizing and structuring data into tables so that information can be stored, managed, and retrieved efficiently. It forms the foundation of the database system and helps maintain data accuracy, consistency, and reliability.

In the **AI-Driven Classroom Engagement Monitoring Using Computer Vision** system, table design is used to store student details, classroom session records, video inputs, behaviour analysis results, engagement reports, and admin information systematically.

Key Components

Tables, Rows, and Columns

- **Tables** store related information such as Student, Video Input, Behaviour Analysis, and Engagement Report.
- **Rows** represent individual records inside a table.
- **Columns** define specific attributes such as `student_id`, `behaviour_type`, `attention_level`, and `report_id`.

Data Types and Column Definitions

Each column is assigned a specific data type such as Integer, Varchar, Float, Date, or DateTime to ensure proper storage and consistency of data.

Primary Keys and Unique Constraints

- **Primary Key (PK)** uniquely identifies each record in a table.
- **Unique Constraints** prevent duplicate values and maintain data integrity.

Example:

- `student_id`
- `report_id`
- `analysis_id`

Foreign Keys and Relationships

Foreign keys are used to connect tables and establish relationships between entities.

Examples:

- session_id links Classroom Session and Video Input.
- video_id links Video Input and Behaviour Analysis.

Normalization

Normalization organizes data into related tables to reduce redundancy and improve database efficiency and consistency.

Indexing

Indexes improve the speed of searching and retrieving records from the database, helping the system process reports and analytics faster.

Tables of the System

Student Table

Field Name	Data Type	Description
student_id	Integer (PK)	Unique ID for student
student_name	Varchar	Name of the student
roll_no	Varchar	Student roll number
class_section	Varchar	Class and section details
email	Varchar	Student email address

AI-DRIVEN CLASSROOM ENGAGEMENT MONITORING SYSTEM

Classroom Session Table

Field Name	Data Type	Description
session_id	Integer (PK)	Unique classroom session ID
session_date	Date	Session date
start_time	Time	Session start time
end_time	Time	Session end time
subject	Varchar	Subject name

Video Input Table

Field Name	Data Type	Description
video_id	Integer (PK)	Unique video ID
file_path	Varchar	Stored video location
video_type	Varchar	Webcam or CCTV
duration	Float	Video duration
session_id	Integer (FK)	Connected classroom session

Behaviour Analysis Table

Field Name	Data Type	Description
analysis_id	Integer (PK)	Unique analysis ID
behaviour_type	Varchar	Detected behaviour
attention_level	Float	Calculated engagement level
timestamp	DateTime	Analysis timestamp
confidence_score	Float	Prediction confidence score

AI-DRIVEN CLASSROOM ENGAGEMENT MONITORING SYSTEM

Engagement Report Table

Field Name	Data Type	Description
report_id	Integer (PK)	Unique report ID
session_date	Date	Report generation date
attentive_percentage	Float	Attentive student percentage
sleepy_percentage	Float	Sleepy student percentage
distracted_percentage	Float	Distracted student percentage
report_file	Varchar	Exported report file

Admin Table

Field Name	Data Type	Description
admin_id	Integer (PK)	Unique admin ID
admin_name	Varchar	Admin name
email	Varchar	Admin email
password	Varchar	Admin password
role	Varchar	System role

Relationship Summary

- One Classroom Session can contain multiple Video Inputs.
- One Video Input can generate multiple Behaviour Analysis records.
- One Engagement Report is generated from Behaviour Analysis data.
- Admin manages reports and monitoring activities.
- Students participate in classroom sessions.

6.4 FUNCTIONAL DIAGRAM

A Functional Diagram represents the major functional modules and operational workflow of the system. It explains how different modules interact with each other to perform classroom engagement monitoring and behavioural analysis.

In the **AI-Driven Classroom Engagement Monitoring Using Computer Vision** system, the Functional Diagram illustrates the complete processing flow starting from video capture to report generation. The system captures classroom video through webcam or CCTV cameras and processes frames using OpenCV and MediaPipe technologies.

The preprocessing module performs frame extraction, face detection, and facial landmark tracking. The behaviour analysis module then identifies student behaviours such as Attentive, Sleepy, Distracted, Talking, Phone Usage, Hand Raise, and Neutral using AI-based classification techniques.

The engagement scoring module calculates classroom engagement levels and generates analytical results. Finally, the reporting module displays real-time monitoring dashboards, behaviour statistics, and downloadable reports for teachers and administrators.

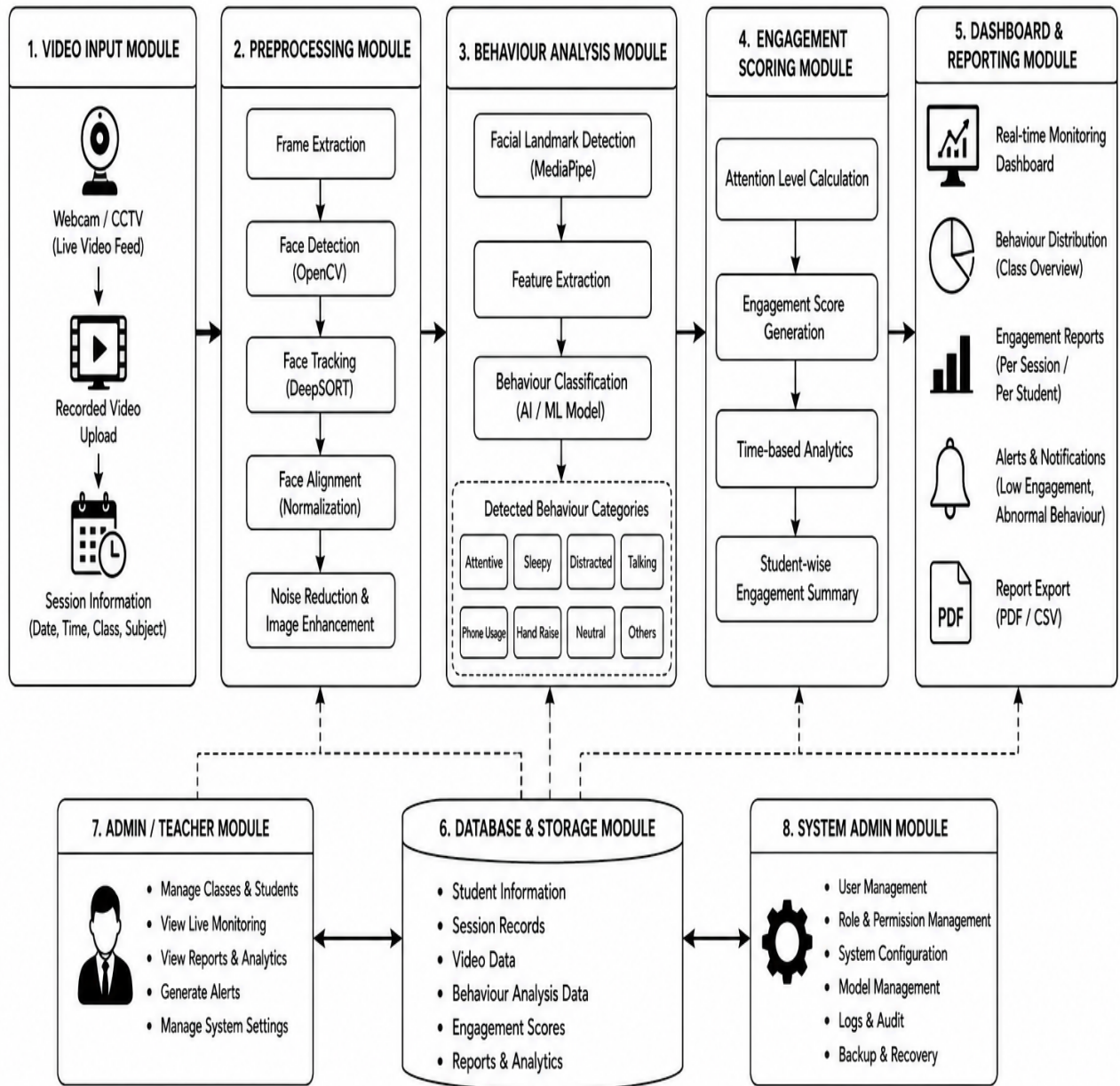
The Functional Diagram helps in understanding the internal working, module interaction, and processing stages of the entire system.

Main Functional Modules

- Video Input Module
- Face Detection Module
- Behaviour Analysis Module
- Engagement Scoring Module
- Database & Storage Module
- Dashboard & Reporting Module

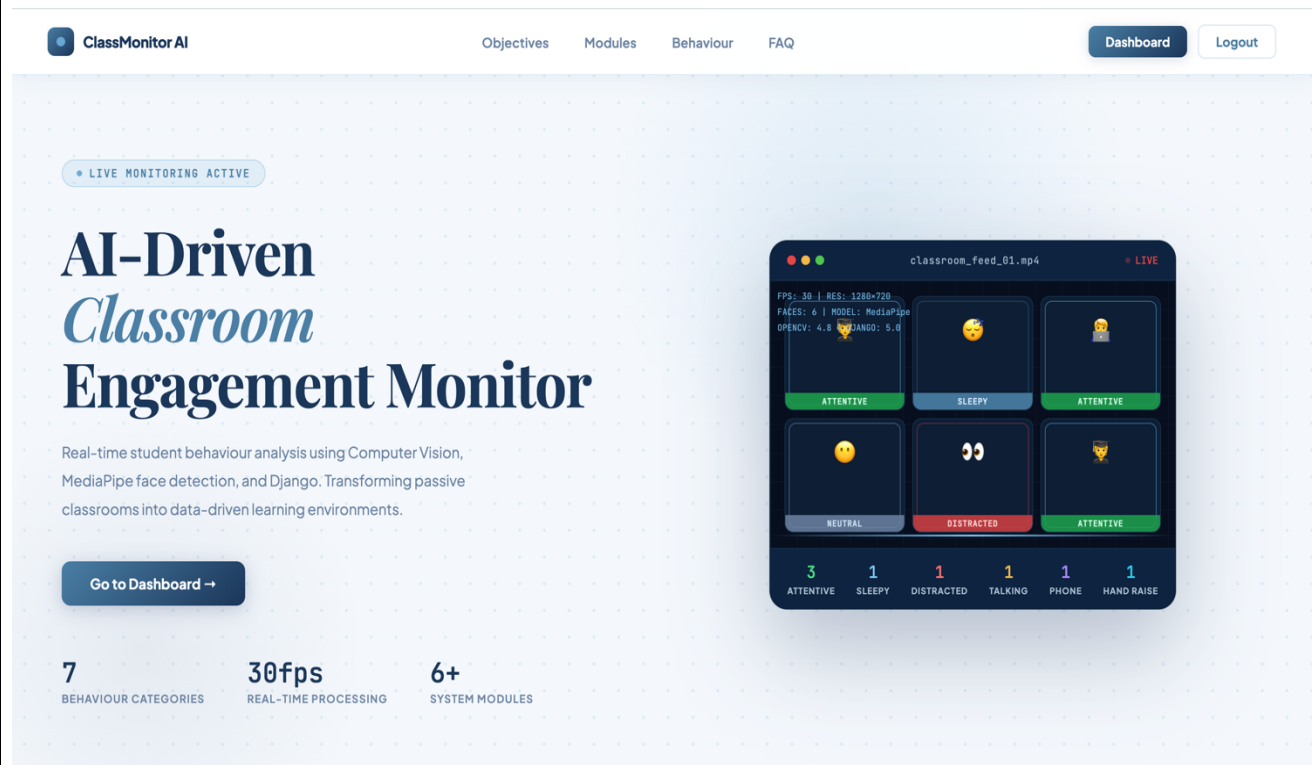
FUNCTIONAL DIAGRAM

AI-DRIVEN CLASSROOM ENGAGEMENT MONITORING USING COMPUTER VISION

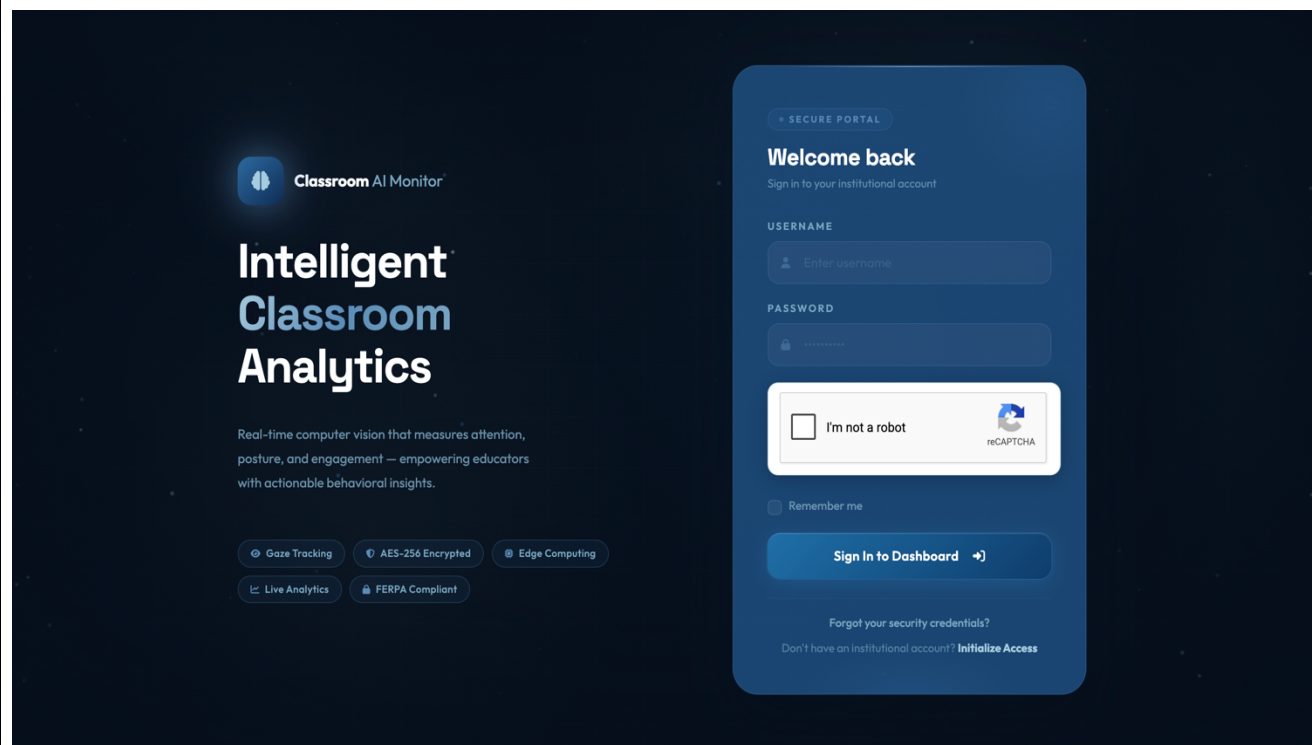


6.5 SNAPSHOTS

Home Page



Login Page



Register Page

 **Classroom AI Monitor**

Initialize Institutional Access

Create your secure educator account and gain access to real-time classroom analytics, behavioral insights, and engagement dashboards.

01 **Create Credentials**
Set up your institutional username and secure password

02 **Verify Identity**
Complete CAPTCHA verification for account security

03 **Access Dashboard**
Start monitoring classroom engagement in real-time

AES-256 Encrypted

FERPA Compliant

Secure Access

Live Analytics

NEW REGISTRATION

Create your account

Join the institutional monitoring platform

USERNAME

Required: 150 characters or fewer. Letters, digits and @/./+/-/_ only.

EMAIL ADDRESS

PASSWORD

At least 8 characters required

CONFIRM PASSWORD

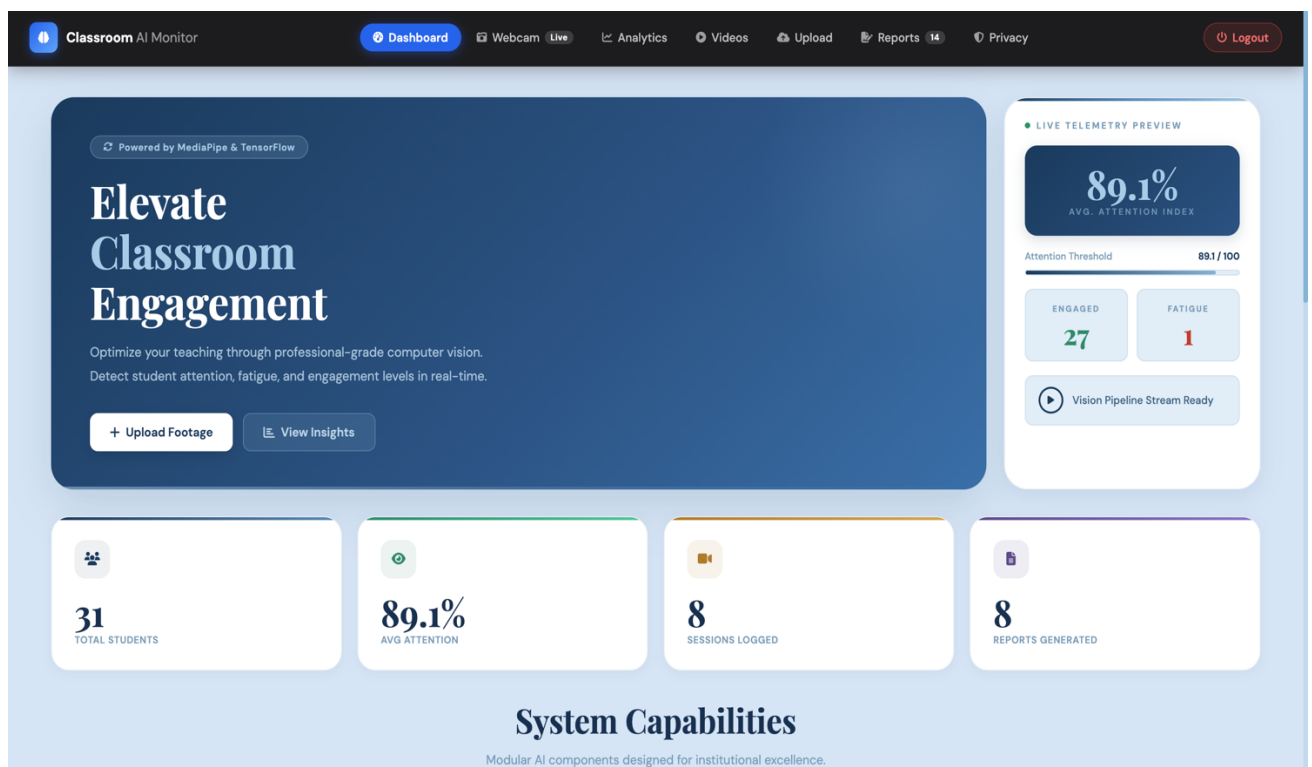
☐ I'm not a robot



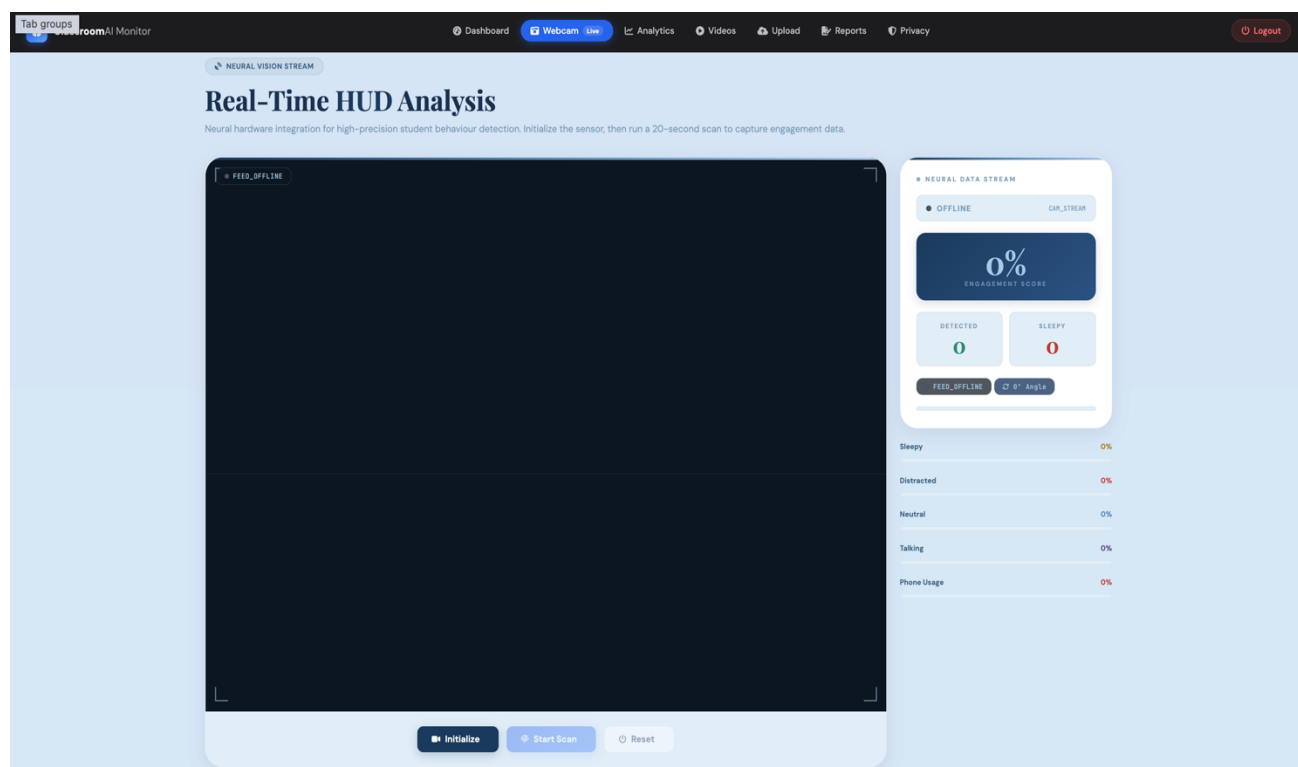
Create Institutional Account

Already have an account? [Login to Console](#)

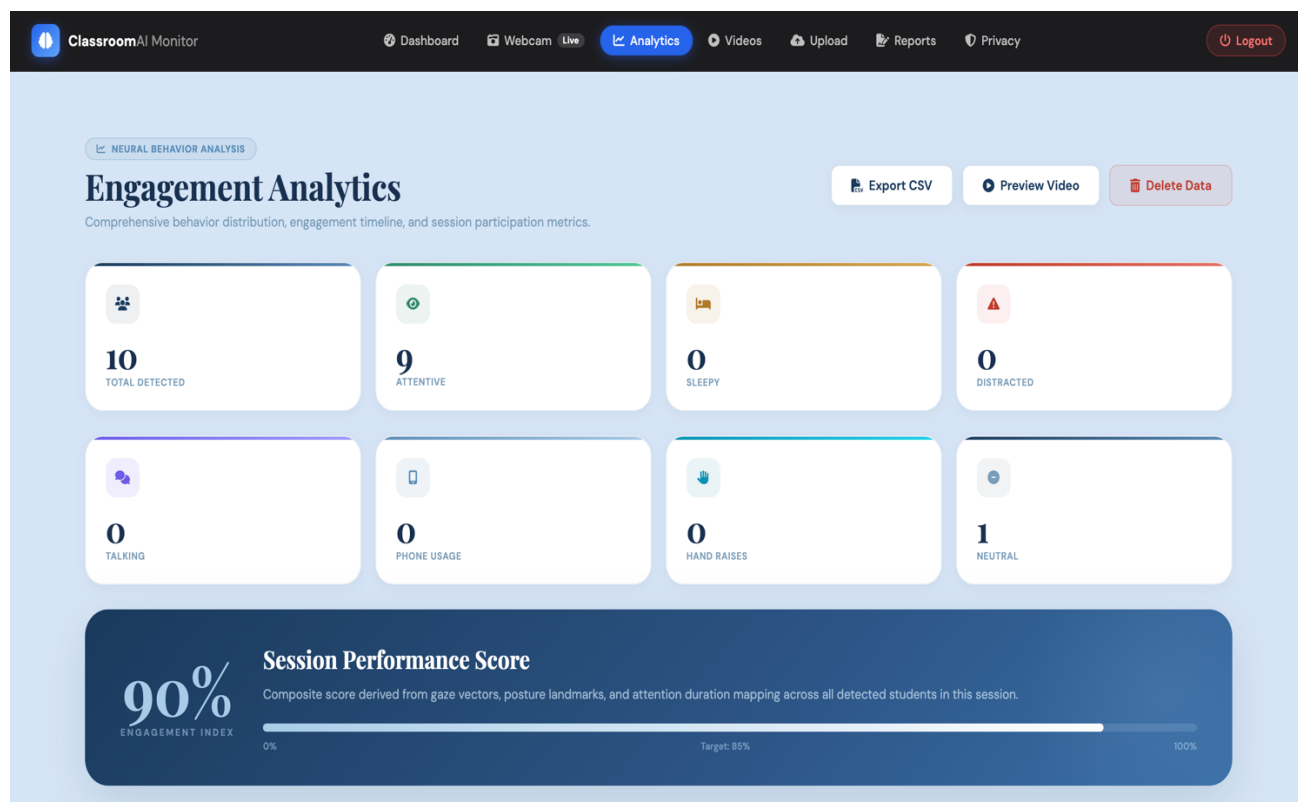
Dashboard Page



Webcam Page



Analytics Page



Video Upload Page

ClassroomAI Monitor

DashboardWebcamLiveAnalyticsVideosUploadReportsPrivacy

Logout

NEURAL REPOSITORY

Upload Session Footage

Submit classroom footage for high-precision behavior analysis.

SESSION REFERENCE TITLE

Enter video title

CAMERA MOUNTING / ORIENTATION

0° (Horizontal)

Select the physical mounting of your camera to ensure AI alignment.

SOURCE VIDEO FILE

Drag & drop or click to browse

Choose fileNo file chosen

Max size: 500MB · Formats: MP4, AVI, MOV

Cancel

Initialize AI Analysis

AI ANALYSIS MODULES

Engagement Matrix

Detecting drift & fatigue levels across the full session timeline.

Accelerated Compute

Optimized local GPU processing — no external data transit.

AES-256 Encryption

All student metadata encrypted at rest and in transit.

PROCESSING PIPELINE

01 Feed Ingestion

Video is ingested locally and queued for neural frame parsing.

02 Neural Mapping

MediaPipe & TensorFlow map facial geometry and posture landmarks per frame.

03 Report Synthesis

Automated generation of engagement summaries and visual dashboards.

Reports Page

tab groups

ClassroomAI Monitor

DashboardWebcamLiveAnalyticsVideosUploadReportsPrivacy

Logout

HISTORICAL REPOSITORY

Export Global Data

Print

Engagement Reports

Neural analysis summaries across all recorded classroom sessions.

8
MANAGED SESSIONS

33%
INSTITUTIONAL AVG

90%
PEAK PERFORMANCE

27
TOTAL OBSERVATIONS

SESSION RECORDS

REFERENCE DATE	TYPE	SESSION CONTEXT	VOLUME	BEHAVIOR DISTRIBUTION	PERFORMANCE INDEX
May 07, 2026	WEBCAM	Live Webcam Scan	Live		0%
May 07, 2026	WEBCAM	Live Webcam Scan	Live		0%
Apr 25, 2026	WEBCAM	Live Webcam Scan	Live		0%

6.6 CODE SNIPPETS

a. Home page

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>AI Classroom Monitor</title>
</head>

<body>

  <!-- Navigation Bar -->
  <nav>
    <h2>ClassMonitor AI</h2>

    <ul>
      <li><a href="#">Home</a></li>
      <li><a href="#">Features</a></li>
      <li><a href="#">Dashboard</a></li>
      <li><a href="#">Login</a></li>
      <li><a href="#">Register</a></li>
    </ul>
  </nav>

  <hr>

  <!-- Hero Section -->
  <section>

    <h1>AI-Driven Classroom Engagement Monitoring System</h1>

    <p>
      This project is designed to monitor classroom engagement using
      Artificial Intelligence, OpenCV, MediaPipe, and Django.
      The system captures live video from webcam or CCTV cameras
      and detects student behaviour in real time.
    </p>

    <p>
      The application helps teachers and institutions identify
      attentive students, sleepy students, distracted behaviour,
      talking activity, phone usage, and overall classroom engagement.
    </p>

    <button>Get Started</button>
    <button>Login</button>
```

</section>

<hr>

<!-- About Section -->

<section>

<h2>About the Project</h2>

<p>

The AI Classroom Monitoring System uses computer vision techniques to analyze facial expressions and engagement levels of students during lectures.

</p>

<p>

The system uses webcam video input and processes frames continuously using OpenCV and MediaPipe Face Mesh.

</p>

<p>

The processed data is displayed on a web dashboard developed using Django framework.

</p>

</section>

<hr>

<!-- Features Section -->

<section>

<h2>System Features</h2>

Real-time face detection

Student behaviour classification

Sleep detection using eye aspect ratio

Head pose estimation

Classroom analytics dashboard

Video upload monitoring

Attendance monitoring

Engagement report generation

</section>

<hr>

<!-- Modules Section -->

<section>

<h2>Core Modules</h2>

<h3>1. Video Capture Module</h3>

<p>

Captures live classroom video using webcam or CCTV camera.

</p>

<h3>2. Face Detection Module</h3>

<p>

Detects faces using MediaPipe face mesh landmarks.

</p>

<h3>3. Behaviour Analysis Module</h3>

<p>

Identifies attentive, distracted, sleepy, and neutral states.

</p>

<h3>4. Dashboard Module</h3>

<p>

Displays classroom engagement statistics and reports.

</p>

</section>

<hr>

<!-- Analytics Section -->

<section>

<h2>Analytics Information</h2>

<table border="1" cellpadding="10">

<tr>

<th>Behaviour</th>

<th>Count</th>

</tr>

<tr>

<td>Attentive</td>

<td>15</td>

</tr>

<tr>

<td>Sleepy</td>

<td>03</td>

</tr>

```
<tr>
  <td>Distracted</td>
  <td>05</td>
</tr>

<tr>
  <td>Neutral</td>
  <td>04</td>
</tr>

</table>

</section>

<hr>

<!-- FAQ Section -->
<section>

  <h2>Frequently Asked Questions</h2>

  <h3>What is the purpose of this project?</h3>
  <p>
    To monitor student engagement using AI and computer vision.
  </p>

  <h3>Which technologies are used?</h3>
  <p>
    Python, Django, OpenCV, MediaPipe, HTML, and JavaScript.
  </p>

  <h3>Is this system real-time?</h3>
  <p>
    Yes, the system processes live video frames continuously.
  </p>

</section>

<hr>

<!-- Contact Section -->
<section>

  <h2>Contact Information</h2>

  <p>Email: classroommonitor@gmail.com</p>

  <p>GitHub: github.com/classroom-monitor</p>
```

<p>LinkedIn: [linkedin.com/in/sadiq](https://www.linkedin.com/in/sadiq)</p>

</section>

<hr>

<!-- Footer -->

<footer>

<p>

Copyright © 2026 AI Classroom Monitoring System

</p>

<p>

Designed and Developed by Sadiq

</p>

</footer>

</body>

</html>

b. Login Page

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Login - Classroom AI Monitor</title>

</head>

<body>

<div>

<!-- LEFT PANEL -->

<div>

<div>

<h2>Classroom AI Monitor</h2>

</div>

<div>

<h1>

Intelligent Classroom Analytics

</h1>

</div>

<p>

Real-time computer vision that measures attention, posture, and engagement — empowering educators with actionable behavioral insights.

</p>

<div>

<p>Gaze Tracking</p>

<p>AES-256 Encrypted</p>

<p>Edge Computing</p>

<p>Live Analytics</p>

<p>FERPA Compliant</p>

</div>

</div>

<!-- LOGIN CARD -->

<div>

<div>

<h3>Secure Portal</h3>

<h1>Welcome back</h1>

<p>Sign in to your institutional account</p>

</div>

{% block auth_content %}

<form method="post" action="{% url 'login' %}">

{% csrf_token %}

<div>

<label for="id_username">Username</label>

<input

type="text"

id="id_username"

name="username"

placeholder="Enter username"

required

>

</div>

<div>

<label for="id_password">Password</label>

<input

type="password"

id="id_password"

name="password"

placeholder="Enter password"

```
        required
    >
</div>

<br>

<div>
    {{ form.captcha }}

    {% if form.captcha.errors %}
        <div>
            {{ form.captcha.errors }}
        </div>
    {% endif %}
</div>

<br>

<div>
    <input type="checkbox" id="rememberMe">
    <label for="rememberMe">Remember me</label>
</div>

<br>

<div>
    <button type="submit">
        Sign In to Dashboard
    </button>
</div>

</form>

<br>

<div>

    <p>
        <a href="{% url 'password_reset' %}">
            Forgot your security credentials?
        </a>
    </p>

    <p>
        Don't have an institutional account?
        <a href="{% url 'register' %}">
            Initialize Access
        </a>
    </p>
```



```
</div>

{% endblock %}

</div>

</div>

</body>
</html>
```

c. Register Page

```
<!DOCTYPE html>
<html>
<head>
<title>Register</title>
</head>
<body>

<h2>User Registration</h2>

<form method="POST">
{% csrf_token %}

<input type="text" name="username" placeholder="Username"><br><br>

<input type="email" name="email" placeholder="Email"><br><br>

<input type="password" name="password" placeholder="Password"><br><br>

<button type="submit">Register</button>

</form>

</body>
</html>
```

Python code

```
from django.contrib.auth.models import User
from django.shortcuts import render
```

```
def register_page(request):
```

```
if request.method == 'POST':
```

```
username = request.POST['username']
email = request.POST['email']
password = request.POST['password']
```

```
User.objects.create_user(
username=username,
email=email,
password=password
)

return render(request, 'register.html')
```

c. Webcam Page

```
{% extends "base.html" %}

{% block title %}Webcam Demo{% endblock %}

{% block content %}

<div class="container">

    <h2>Real-Time Classroom AI Monitor</h2>
    <p>Analyze student engagement using webcam.</p>

    <!-- Webcam Section -->
    <div class="camera-box">
        <video id="webcam" autoplay playsinline></video>
        <canvas id="canvas-overlay"></canvas>
    </div>

    <!-- Controls -->
    <div class="controls">
        <button id="start-btn">Initialize Camera</button>
        <button id="record-btn" disabled>Start Scan</button>
        <button id="stop-btn" disabled>Stop</button>
    </div>

    <!-- Stats -->
    <div class="stats-box">
        <h3>Live Statistics</h3>

        <p>Engagement Score: <span id="score-val">0%</span></p>
        <p>Detected Faces: <span id="face-count">0</span></p>
        <p>Sleepy Students: <span id="sleepy-count">0</span></p>

        <p>Attentive: <span id="attentive-val">0%</span></p>
        <p>Distracted: <span id="distracted-val">0%</span></p>
        <p>Talking: <span id="talking-val">0%</span></p>
        <p>Phone Usage: <span id="phone-val">0%</span></p>

        <p id="state-val">Waiting for signal...</p>
    </div>

</div>
```

```
<!-- Result -->
<div id="result-card" style="display:none;">
  <h3>Session Complete</h3>

  <p>Final Engagement: <span id="final-score">0%</span></p>

  <p id="analysis-summary"></p>

  <button id="reset-btn">New Session</button>
</div>
```

```
</div>
```

```
{% endblock %}
```

Python code

```
import cv2
from django.http import StreamingHttpResponse
from django.shortcuts import render

camera = cv2.VideoCapture(0)
```

```
def webcam_page(request):
    return render(request, 'webcam.html')
```

```
def generate_frames():
```

```
    while True:
```

```
        success, frame = camera.read()
```

```
        if not success:
            break
```

```
        ret, buffer = cv2.imencode('.jpg', frame)
```

```
        frame = buffer.tobytes()
```

```
        yield (b'--frame\r\n'
               b'Content-Type: image/jpeg\r\n\r\n' +
               frame + b'\r\n')
```

```
def video_feed(request):
```

```
    return StreamingHttpResponse(
        generate_frames(),
        content_type='multipart/x-mixed-replace; boundary=frame')
```

d. video upload page

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Upload Video{% endblock %}

{% block content %}
<div class="container mt-4">

    <h2>Upload Classroom Video</h2>
    <p>Upload video for AI engagement analysis.</p>

    <form method="post" enctype="multipart/form-data">
        {% csrf_token %}

        <div class="mb-3">
            <label>Session Title</label>
            {{ form.title }}
        </div>

        <div class="mb-3">
            <label>Camera Angle</label>
            {{ form.camera_angle }}
        </div>

        <div class="mb-3">
            <label>Video File</label>
            {{ form.video_file }}
        </div>

        <button type="submit" class="btn btn-primary">
            Upload & Analyze
        </button>

        <a href="{% url 'videos:video_list' %}" class="btn btn-secondary">
            Cancel
        </a>
    </form>

</div>
{% endblock %}
```

Python code

```
from django.shortcuts import render
from django.core.files.storage import FileSystemStorage
```

```
def upload_video(request):
```

```
    if request.method == 'POST':
```

```
if request.FILES.get('video'):
```

```
video = request.FILES['video']
```

```
fs = FileSystemStorage()
```

```
fs.save(video.name, video)
```

```
return render(request, 'upload.html')
```

e. Analytics Page

```
{% extends "base.html" %}
```

```
{% block title %}Analytics{% endblock %}
```

```
{% block content %}
```

```
<h2>{{ video_title|default:"Analytics Dashboard" }}</h2>
```

```
{% if message %}
```

```
<p>{{ message }}</p>
```

```
<a href="{% url 'videos:video_upload' %}">Upload Video</a>
```

```
{% endif %}
```

```
<hr>
```

```
<h3>Student Analytics</h3>
```

```
<ul>
```

```
<li>Total Students : {{ total_students }}</li>
```

```
<li>Attentive : {{ attentive }}</li>
```

```
<li>Sleepy : {{ sleepy }}</li>
```

```
<li>Distracted : {{ distracted }}</li>
```

```
<li>Talking : {{ talking|default:"0" }}</li>
```

```
<li>Phone Usage : {{ phone|default:"0" }}</li>
```

```
<li>Hand Raises : {{ hand_raises|default:"0" }}</li>
```

```
<li>Neutral : {{ neutral|default:"0" }}</li>
```

```
</ul>
```

```
<hr>
```

```
<h3>Engagement Score</h3>
```

```
<p>{{ engagement_percentage }}%</p>
```

```
<hr>
```

```
{% if data_type == 'video' and video %}
```

```
<a href="{% url 'videos:video_export_csv' video.pk %}">Export CSV</a>
```


Preview Video

<form method="POST" action="{% url 'videos:video_delete' video.pk %}">
 {% csrf_token %}
 <button type="submit">Delete Data</button>
</form>
{% endif %}

<hr>

<h3>Behavior Summary</h3>

<table border="1" cellpadding="10">
 <tr>
 <th>Behavior</th>
 <th>Count</th>
 </tr>
 <tr>
 <td>Attentive</td>
 <td>{{ attentive }}</td>
 </tr>
 <tr>
 <td>Sleepy</td>
 <td>{{ sleepy }}</td>
 </tr>
 <tr>
 <td>Distracted</td>
 <td>{{ distracted }}</td>
 </tr>
 <tr>
 <td>Talking</td>
 <td>{{ talking|default:"0" }}</td>
 </tr>
 <tr>
 <td>Phone Usage</td>
 <td>{{ phone|default:"0" }}</td>
 </tr>
 <tr>
 <td>Hand Raises</td>
 <td>{{ hand_raises|default:"0" }}</td>
 </tr>
 <tr>
 <td>Neutral</td>
 <td>{{ neutral|default:"0" }}</td>
 </tr>
</table>

<hr>

```
{% if data_type == 'video' and video and video.heatmap_image %}
  <h3>Attention Heatmap</h3>
  
{% endif %}
```

```
{% endblock %}
```

Python code

```
import cv2
```

```
face_cascade = cv2.CascadeClassifier(
cv2.data.harcascades +
'haarcascade_frontalface_default.xml'
)
camera = cv2.VideoCapture(0)
```

```
while True:
```

```
    success, frame = camera.read()
```

```
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
```

```
    faces = face_cascade.detectMultiScale(gray, 1.3, 5)
```

```
    for (x, y, w, h) in faces:
```

```
        cv2.rectangle(frame,
        (x, y),
        (x+w, y+h),
        (255, 0, 0),
        2)
```

```
cv2.imshow('Face Detection', frame)
```

```
if cv2.waitKey(1) == 27:
```

```
    break
```

```
<canvas id="myChart"></canvas>
```

```
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
```

```
<script>
```

```
const ctx = document.getElementById('myChart');
```

```
new Chart(ctx, {
type: 'bar',
data: {
labels: ['Attentive', 'Sleepy', 'Distracted'],
```

```
datasets: [{  
  label: 'Students',  
  data: [30, 5, 10],  
  borderWidth: 1  
}]  
{  
});  
</script>
```

f. Reports Page

```
{% extends "base.html" %}  
  
{% block title %}Reports{% endblock %}  
  
{% block content %}  
  
<h2>Engagement Reports</h2>  
  
<p>Historical classroom engagement reports and analytics.</p>  
  
<hr>  
  
<h3>Summary</h3>  
  
<ul>  
  <li>Total Sessions : {{ reports|length }}</li>  
  <li>Average Engagement : {{ avg_engagement }}%</li>  
  <li>Best Engagement : {{ best_engagement }}%</li>  
  <li>Total Students : {{ total_students_stat }}</li>  
</ul>  
  
<hr>  
  
<a href="{% url 'reports_export_csv' %}">Export CSV</a>  
  
<br><br>  
  
<button onclick="window.print()">Print</button>  
  
<hr>  
  
<table border="1" cellpadding="10">  
  <thead>  
    <tr>  
      <th>Date</th>  
      <th>Type</th>  
      <th>Title</th>  
      <th>Total</th>
```



```

        <th>Attentive</th>
        <th>Sleepy</th>
        <th>Distracted</th>
        <th>Talking</th>
        <th>Phone</th>
        <th>Hand Raises</th>
        <th>Engagement</th>
        <th>Actions</th>
    </tr>
</thead>

<tbody>
    {% for report in reports %}
    <tr>
        <td>{{ report.date|date:"M d, Y" }}</td>

        <td>
            {% if report.type == 'video' %}
                Video
            {% else %}
                Webcam
            {% endif %}
        </td>

        <td>{{ report.title }}</td>

        <td>{{ report.total }}</td>

        <td>{{ report.attentive }}</td>

        <td>{{ report.sleepy }}</td>

        <td>{{ report.distracted }}</td>

        <td>{{ report.talking }}</td>

        <td>{{ report.phone_usage }}</td>

        <td>{{ report.hand_raises }}</td>

        <td>{{ report.engagement }}%</td>

        <td>
            {% if report.type == 'video' and report.heatmap %}
                <a href="{{ report.heatmap }}">Heatmap</a>
                <br><br>
            {% endif %}

            <a href="{% url 'analytics_detail' report.id %}?type={{ report.type }}">
                Analytics
    
```

```
</a>

<br><br>

<form method="POST" action="{% url 'report_delete' report.id %}">
    {% csrf_token %}
    <button type="submit">Delete</button>
</form>
</td>
</tr>

{% empty %}
<tr>
    <td colspan="12">
        No reports available
    </td>
</tr>
{% endfor %}
</tbody>
</table>

{% endblock %}

python code

from django.shortcuts import render

def reports(request):

    data = {
        'attentive': 30,
        'sleepy': 5,
        'distracted': 10,
    }

    return render(request, 'reports.html', data)
```

7.1 System Implementation

The implementation phase is one of the most important stages in the development of the **AI-Driven Classroom Engagement Monitoring Using Computer Vision** system. In this phase, all designed modules, AI models, frontend interfaces, and database functionalities were integrated into a working application.

The system was developed using Python and Django framework for backend processing and web application management. OpenCV was used for real-time video processing, while MediaPipe FaceMesh was implemented for facial landmark detection and tracking. The frontend interface was developed using HTML, CSS, and JavaScript to provide a responsive and user-friendly dashboard.

The implementation process started with video capture integration using webcam and CCTV feeds. Frames from the video stream were processed continuously using OpenCV. The MediaPipe FaceMesh model detected facial landmarks for multiple faces in real time. The system then analyzed facial movements, eye aspect ratio (EAR), mouth aspect ratio (MAR), and head pose estimation to identify student behaviour patterns.

The AI classification module categorized behaviours into:

- Attentive
- Sleepy
- Distracted
- Talking
- Phone Usage
- Hand Raise
- Neutral

The engagement analysis results were displayed on the dashboard using charts, statistics, and behavioural summaries. All engagement records and reports were stored in the database using Django ORM and SQLite/PostgreSQL database integration.

The system was tested using classroom videos and live webcam feeds to evaluate real-time performance and detection accuracy.

7.2 Modules Implemented

The following modules were successfully implemented in the project:

1. User Authentication Module

Handles login, registration, and secure access to the system.

2. Video Capture Module

Captures live classroom video using webcam or CCTV input or video can be upload.

3. Face Detection Module

Detects and tracks multiple student faces using MediaPipe FaceMesh.

4. Behaviour Analysis Module

Classifies student behaviour using facial landmarks and AI logic.

5. Dashboard Module

Displays engagement analytics, charts, and classroom statistics.

6. Report Generation Module

Generates downloadable engagement reports in CSV format.

7. Database Management Module

Stores classroom sessions, reports, and behavioural records.

7.3 System Testing

System testing is performed to ensure that the application functions correctly and meets project requirements. The testing phase helps identify errors, improve system reliability, and validate the behaviour analysis process.

Different types of testing were conducted during project development.

1. Functional Testing

Functional testing verifies whether all modules perform their intended operations correctly.

Module	Expected Result	Status
Login System	User authentication successful	Passed
Video Capture	Webcam feed displayed correctly	Passed
Face Detection	Multiple faces detected	Passed
Behaviour Analysis	Behaviour classification generated	Passed
Dashboard	Reports and charts displayed	Passed
Report Export	CSV report generated	Passed

2. Performance Testing

Performance testing evaluates the speed and efficiency of the system during real-time execution.

Parameter	Result
Frame Processing Speed	Approx. 30 FPS
Face Detection Time	Real-Time
Behaviour Classification	Under 33ms
Dashboard Loading	Fast

3. Accuracy Testing

The behaviour analysis model was tested under different classroom conditions such as lighting changes, multiple students, and varying head movements.

Behaviour	Detection Accuracy
Attentive	High
Sleepy	Moderate to High
Distracted	Moderate
Talking	Moderate
Phone Usage	Moderate
Hand Raise	Moderate
Neutral	High

4. User Interface Testing

The web dashboard and monitoring pages were tested for usability, responsiveness, and navigation.

Results:

- Easy navigation
- Responsive layout
- Real-time analytics display
- User-friendly interface

Merits

1. Automated Monitoring

The system automatically monitors classroom engagement without manual observation.

2. Real-Time Analysis

Behaviour analysis is performed instantly using live video processing.

3. Improved Teaching Support

Teachers can identify engagement issues and improve classroom interaction.

4. Multi-Face Detection

The system can detect and analyze multiple students simultaneously.

5. Data-Driven Insights

Provides reports, statistics, and engagement analytics for academic improvement.

6. Web-Based Dashboard

Accessible through a browser with centralized monitoring and reporting.

Demerits

1. Lighting Dependency

Performance may reduce in low-light or overexposed environments.

2. Occlusion Issues

Faces blocked by objects or students may not be detected accurately.

3. Hardware Requirement

Real-time video processing requires good system performance.

4. Behaviour Accuracy Limitations

Some behaviours may overlap and reduce classification accuracy.

5. Privacy Concerns

Classroom video monitoring requires proper data handling and permissions.

9.1 FUTURE ENHANCEMENTS

The project can be further improved by adding advanced AI and analytics features.

1. Emotion Recognition

Future versions can detect emotions such as confusion, happiness, frustration, or stress.

2. Per-Student Analysis

Individual engagement tracking can be added for detailed student performance monitoring.

3. Mobile Application

A mobile app can provide live monitoring and notifications for teachers.

4. Cloud Deployment

The system can be deployed on cloud platforms for institutional scalability.

5. LMS Integration

Integration with Moodle, Google Classroom, or Canvas can improve educational analytics.

6. Multi-Camera Support

Multiple camera feeds can reduce occlusion and improve monitoring coverage.

7. AI Model Optimization

Advanced deep learning models can improve behaviour detection accuracy.

8. Audio Analysis

Voice participation and classroom interaction can be analyzed along with video monitoring.

Conclusion

The **AI-Driven Classroom Engagement Monitoring Using Computer Vision** system successfully demonstrates the application of Artificial Intelligence and Computer Vision technologies in modern education environments.

The project provides an automated solution for monitoring classroom engagement using real-time video analysis. By integrating OpenCV, MediaPipe, Django, and AI-based behaviour analysis techniques, the system can detect important classroom behaviours such as attentiveness, sleepiness, distraction, talking, phone usage, hand raise, and neutral behaviour.

The system helps teachers and institutions understand classroom interaction patterns through analytical dashboards, reports, and engagement statistics. It reduces manual observation workload and provides objective, data-driven insights for improving teaching quality and student participation.

Although the system has some limitations related to lighting conditions and behavioural accuracy, it establishes a strong foundation for future AI-based educational monitoring systems.

Overall, the project demonstrates how Computer Vision and Artificial Intelligence can enhance smart classroom environments and support data-driven educational improvement.

REFERENCE & BIBLIOGRAPHY

References Summary Table

The following references were used in the development of the project titled “**AI-Driven Classroom Engagement Monitoring Using Computer Vision**”:

Official Documentation

- OpenCV Documentation.
Available at: [OpenCV Official Website](#)
- Google Documentation.
Available at: [MediaPipe Documentation](#)
- Django Software Foundation Documentation.
Available at: [Django Documentation](#)
- NumPy Documentation.
Available at: [NumPy Documentation](#)

Web Development Resources

- W3Schools
Available at: [W3Schools](#)
- Mozilla
Available at: [MDN Web Docs](#)

Libraries & Tools

- Chart.js Documentation.
Available at: [Chart.js Official Website](#)
- Visual Studio Code
Available at: [Visual Studio Code](#)
- GitHub
Available at: [GitHub](#)

AI-DRIVEN CLASSROOM ENGAGEMENT MONITORING SYSTEM

Online Tutorials & Research

- Various YouTube tutorials on OpenCV, MediaPipe, and Django development.

Category	Resource	Description / Availability
Official Documentation	OpenCV	Official documentation and website for computer vision library
Official Documentation	MediaPipe	Official documentation for face mesh and vision tools
Official Documentation	Django	Official framework documentation for backend development
Official Documentation	NumPy	Official documentation for numerical computing library
Web Development Resources	W3Schools	Beginner-friendly tutorials and examples for web technologies
Web Development Resources	MDN Web Docs	Reference guides and standards-based web documentation
Libraries & Tools	Chart.js	Documentation for charting and data visualization library
Libraries & Tools	Visual Studio Code	Development environment and editor documentation
Libraries & Tools	GitHub	Version control, repository hosting, and collaboration platform
Online Tutorials & Research	YouTube Tutorials	Video tutorials on OpenCV, MediaPipe, and Django development